

Rangkuman Step Perpanjangan SSL

Domain: orlansoftarbechem.com

Tanggal: 10 Juni 2026

Status Awal

SSL pada domain `orlansoftarbechem.com` terdeteksi expired, sehingga akses HTTPS menampilkan warning/error certificate. Aplikasi ERP masih aktif dan dapat diakses, namun sertifikat SSL pada Tomcat sudah melewati masa berlaku.

Step Pekerjaan yang Dilakukan

1. Pengecekan DNS domain

```
nslookup orlansoftarbechem.com
```

Hasil domain mengarah ke IP server:

```
187.77.113.16
```

2. Pengecekan akses aplikasi

```
curl -I http://orlansoftarbechem.com  
curl -I https://orlansoftarbechem.com
```

Hasil:

- HTTP normal.
- HTTPS error karena SSL expired.

3. Pengecekan port di server

```
ss -tulpen | grep -E ':80|:443|:8080|:8443'
```

Hasil:

- Port 80 digunakan oleh Java/Tomcat.
- Port 443 digunakan oleh Java/Tomcat.

Kesimpulan: SSL tidak menggunakan Nginx, tetapi langsung dari Tomcat.

4. Pengecekan konfigurasi SSL Tomcat

Dilakukan pengecekan pada file:

```
/opt/orlansoft_erp/tomcat/conf/server.xml
```

Ditemukan bahwa SSL menggunakan file JKS:

```
/opt/orlansoft_erp/tomcat/conf/arbechemkeystore2025.jks
```

5. Pengecekan masa berlaku certificate lama

```
/opt/orlansoft_erp/openjdk/bin/keytool -list -v  
-keystore /opt/orlansoft_erp/tomcat/conf/arbechemkeystore2025.jks  
-alias tomcat
```

Hasil:

- Certificate lama sudah expired pada 12 Mei 2026.

6. Backup file lama

```
cp /opt/orlansoft_erp/tomcat/conf/server.xml /opt/orlansoft_erp/tomcat/conf/  
server.xml.bak_ssl_20260609  
  
cp /opt/orlansoft_erp/tomcat/conf/arbechemkeystore2025.jks /opt/orlansoft_erp/  
tomcat/conf/arbechemkeystore2025.jks.expired_20260609
```

7. Install Certbot

```
apt update  
apt install certbot -y
```

8. Stop service ERP sementara

```
systemctl stop OrlansoftERP.service
```

Lalu dipastikan port 80 dan 443 sudah kosong:

```
ss -tulpen | grep -E ' :80| :443 '
```

9. Generate SSL baru Let's Encrypt

```
certbot certonly --standalone -d orlansoftarbechem.com
```

SSL baru berhasil dibuat di lokasi:

```
/etc/letsencrypt/live/orlansoftarbechem.com/
```

10. Convert SSL ke format PKCS12

```
openssl pkcs12 -export  
-out /tmp/orlansoftarbechem.p12  
-inkey /etc/letsencrypt/live/orlansoftarbechem.com/privkey.pem  
-in /etc/letsencrypt/live/orlansoftarbechem.com/cert.pem  
-certfile /etc/letsencrypt/live/orlansoftarbechem.com/chain.pem  
-name tomcat
```

11. Convert PKCS12 ke JKS Tomcat

```
/opt/orlansoft_erp/openjdk/bin/keytool -importkeystore  
-srckeystore /tmp/orlansoftarbechem.p12  
-srcstoretype PKCS12  
-destkeystore /tmp/arbechemkeystore2026.jks  
-deststoretype JKS  
-srcalias tomcat  
-destalias tomcat  
-noprompt
```

12. Validasi JKS baru

```
/opt/orlansoft_erp/openjdk/bin/keytool -list -v  
-keystore /tmp/arbechemkeystore2026.jks  
-alias tomcat
```

Hasil SSL baru:

```
Issuer      : Let's Encrypt YR2  
Valid To    : 08 September 2026
```

13. Replace keystore lama dengan keystore baru

```
cp /tmp/arbechemkeystore2026.jks /opt/orlansoft_erp/tomcat/conf/  
arbechemkeystore2025.jks  
chmod 600 /opt/orlansoft_erp/tomcat/conf/arbechemkeystore2025.jks
```

14. Start kembali service ERP

```
systemctl start OrlansoftERP.service
```

Cek status service:

```
systemctl status OrlansoftERP.service
```

Hasil:

```
Active: active (running)
```

15. Validasi akses HTTPS

```
curl -I https://orlansoftarbechem.com
```

Hasil:

```
HTTP/1.1 200
```

16. Validasi certificate aktif di website

```
echo | openssl s_client -connect orlansoftarbechem.com:443 -servername  
orlansoftarbechem.com 2>/dev/null | openssl x509 -noout -subject -issuer -dates
```

Hasil:

```
subject=CN = orlansoftarbechem.com
issuer=C = US, O = Let's Encrypt, CN = YR2
notBefore=Jun 10 04:03:31 2026 GMT
notAfter=Sep 8 04:03:30 2026 GMT
```

Setup Auto Renew SSL

Agar SSL tidak expired kembali, dibuat script auto-renew:

```
/usr/local/sbin/renew_orlansoftarbechem_ssl.sh
```

Log script:

```
/var/log/renew_orlansoftarbechem_ssl.log
```

Jadwal cron:

```
00 01 * * 0 /usr/local/sbin/renew_orlansoftarbechem_ssl.sh >/dev/null 2>&1
```

Artinya setiap hari Minggu jam 01:00 server akan mengecek SSL. Jika SSL masih lebih dari 21 hari, script tidak melakukan renew dan tidak restart ERP. Jika SSL sudah mendekati expired, script akan renew SSL, convert ke JKS, replace keystore Tomcat, lalu restart ERP.

Hasil Akhir

Perpanjangan SSL berhasil dilakukan.

```
Domain       : orlansoftarbechem.com
HTTPS        : Normal
Status       : HTTP/1.1 200
Service ERP  : Active / Running
SSL          : Let's Encrypt
Expired      : 08 September 2026
Auto Renew   : Sudah dijadwalkan setiap Minggu jam 01:00
```